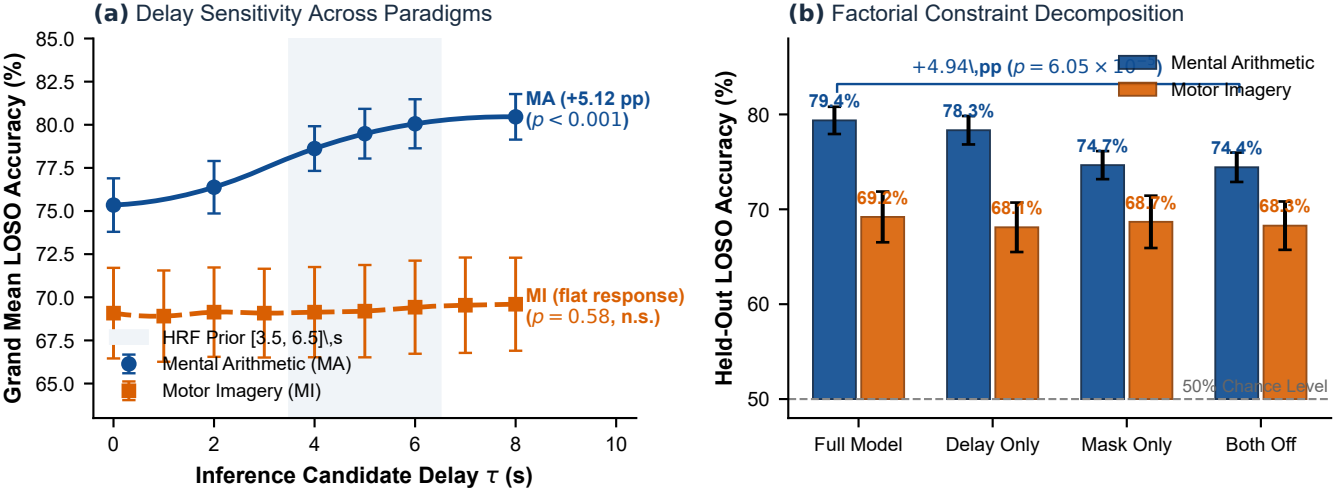

(a) Delay Sensitivity Across Paradigms
Grand Mean LOSO Accuracy (%)
85.0
82.5
80.0
77.5
75.0
72.5
70.0
67.5
65.0
0
2
4
6
8
10
Inference Candidate Delay τ (s)
MA (+5.12 pp)
(p < 0.001)
MI (flat response)
(p = 0.58, n.s.)
HRF Prior [3.5, 6.5]\,s
Mental Arithmetic (MA)
Motor Imagery (MI)
(b) Factorial Constraint Decomposition
Held-Out LOSO Accuracy (%)
80
70
60
50
+4.94\,pp (p = 6.05 × 10
Mental Arithmetic
Motor Imagery
79.4%
69.2%
78.3%
68.1%
74.7%
68.7%
74.4%
68.3%
50% Chance Level
Full Model
Delay Only
Mask Only
Both Off